

Human Activity Recognition (HAR) – Battery-Efficient Personalized Model

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Abstract: Human Activity Recognition (HAR) is a key technology for health, fitness, and context-aware mobile apps. However, deploying precise models on devices with limited battery life remains challenging. This research looks into an efficient, personalized HAR method that relies solely on essential acceleration features. This approach significantly reduces computational demands and saves energy. By training individual models for each user and comparing them to a general model, we demonstrate a clear improvement in recognition accuracy while maintaining low battery usage. Our method achieves, on average, 7% higher accuracy than standard general models. It also shows the potential of adaptable sensor techniques for practical use in mobile and wearable devices.

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I. INTRODUCTION (

In recent years, improvements in wearable technology and smart devices have opened up new opportunities for recognizing human activities from sensor data. Human Activity Recognition (HAR) systems are crucial in healthcare, fitness tracking, and context-aware computing. However, using these systems on battery-operated mobile devices is still challenging due to limited computing power and energy limits.

This study provides a practical solution by focusing on two main areas: selecting only essential sensor features that need little computation and customizing models to fit each user's specific movement patterns. We develop lightweight classifiers designed for individual users, using carefully chosen low-power features instead of complex, generalized models.

Our approach not only cuts down the system's energy consumption but also significantly boosts recognition accuracy.

Our thorough experiments demonstrate that personalized models greatly outperform general models, achieving higher precision and reliability in real-world applications. This energy-efficient method makes the technology more suitable for long-term use in wearable and mobile applications and paves the way for future advancements like adaptive sensor sampling and improved resource management.

II. LITERATURE REVIEW

Human Activity Recognition (HAR) has become a key area of research for applications such as healthcare, fitness

tracking, and intelligent mobile systems. HAR involves recognizing human actions from sensor data using machine learning models. However, deploying such systems on battery-constrained devices poses challenges in computation and power consumption [1], [2].

A. HAR Model Design and Feature Selection

Traditional HAR models depend on handcrafted statistical features, including mean, standard deviation, correlation, and entropy. These features come from accelerometer and gyroscope signals [3], [4]. They are usually applied in classical classifiers such as Support Vector Machine (SVM), Decision Tree (DT), and Random Forest (RF). However, increasing the number of features raises computational and energy demands.

To address this issue, recent studies have focused on selecting features and reducing dimensionality using methods like Linear Discriminant Analysis (LDA), Principal Component Analysis (PCA), and genetic algorithms [5], [6]. These approaches improve model efficiency by choosing the most relevant features, which allows for real-time recognition on low-power devices.

Additionally, hybrid deep learning models, such as CNNs, LSTMs, and CNN-LSTM combinations, effectively capture spatial-temporal dependencies [7], [8]. However, these models can be computationally intensive. As a result, lightweight models that focus only on essential acceleration features are now widely researched [9].

B. Energy Efficiency and Battery-Aware Techniques

Energy efficiency is critical for HAR systems implemented on mobile and wearable platforms. Frequent sensor sampling and continuous computation drain device batteries rapidly. Adaptive sampling has been proposed to address this by adjusting the sampling rate based on detected activity levels [10], [11].

For instance, activity-aware sampling dynamically reduces sampling rates during static activities like sitting and increases them during dynamic movements such as running. This adaptive strategy achieves up to 56%–70% energy savings with minimal accuracy degradation [12]. Another line of research focuses on sensor reduction, using only accelerometer data instead of multiple sensors to further reduce energy consumption [13]. These methods strike a balance between energy efficiency and accuracy for long-term wearable applications.

C. Personalization and Its Role in Accuracy Improvement

HAR accuracy often drops when a generalized model is applied to different users due to variations in movement style, body type, and device position [14]. Personalized HAR models, fine-tuned using a small subset of user-specific samples, significantly improve performance.

Studies such as Weiss and Lockhart [15] demonstrate that user-adaptive models achieve 5–10% higher accuracy compared to general ones. More recently, meta-learning and transfer learning have enabled lightweight personalization through selective fine-tuning of parameters rather than retraining entire models [16], [17]. These approaches enhance precision while keeping computation and energy consumption low—key for real-world deployment.

D. Machine Learning Models and Computational Cost

Machine learning models vary in complexity and energy demand. Logistic Regression, SVM, and Random Forest provide fast inference and low computational overhead but depend on manual feature engineering [18]. Deep learning models such as CNN and LSTM deliver superior accuracy but require more memory and processing time [19]. Hybrid CNN–LSTM models integrate both spatial and temporal dependencies but often exceed the computational limits of mobile processors [20].

Therefore, lightweight classifiers optimized with minimal features, such as LDA or logistic regression, are more practical for embedded systems, balancing recognition accuracy and computational efficiency.

E. Energy-Saving Mechanisms in Mobile and Wearable Devices

To further reduce power usage, researchers have proposed adaptive sensor sampling, duty cycling, and on-device edge processing [11], [12]. These approaches dynamically manage sensor activation and minimize wireless data transfer.

Feature reduction using LDA and PCA further decreases the data volume processed by classifiers [5]. Compressive sensing-based techniques have also been explored to reconstruct and classify signals directly from fewer samples, saving approximately 20% energy with 88% recognition accuracy [21].

F. Identified Research Gaps and Contributions

Despite significant advances, two major gaps persist: (1) achieving high accuracy while maintaining energy efficiency, and (2) incorporating user personalization into real-world, low-power HAR systems.

This research addresses these gaps by proposing a personalized, lightweight HAR framework that utilizes only essential acceleration features to minimize computation. The system achieves an average 7% improvement in recognition accuracy over general models while reducing battery usage. Furthermore, it integrates adaptive sensing strategies to make HAR systems more practical for mobile and wearable devices.

III. METHODS

A. Dataset Description

Our experiments utilize the widely recognized **UCI Human Activity Recognition Using Smartphones Dataset (UCI HAR Dataset)**, created by Anguita et al. This dataset comprises sensor data collected from 30 participants over various common daily activities captured by smartphones' built-in inertial sensors.

Each participant performed six activities:

- I. Walking (forward in a straight line)
- II. Walking Upstairs
- III. Walking Downstairs
- IV. Sitting
- V. Standing
- VI. Laying (lying down or sleeping)

The data consists of tri-axial accelerometer and gyroscope readings recorded at a sampling rate of 50 Hz, resulting in 561 features extracted per sample. The dataset is partitioned into separate training and testing sets, with 7352 and 2947 samples respectively, properly labeled with activity type and subject identifiers.

This dataset enables robust evaluation of activity recognition algorithms and has become a benchmark in the field due to its well-structured design, balance of activities, and realistic sensor data variability.

B. Data Preprocessing

In this study, preprocessing plays a critical role in preparing the raw sensor data for effective model training and evaluation. Initially, the separate training and testing datasets from the UCI HAR dataset were combined into a single dataset for unified processing. Each record in this dataset includes tri-axial accelerometer and gyroscope readings, subject identifiers, and activity labels corresponding to six common human activities.

To handle any missing or irregular data points, appropriate data cleaning techniques such as interpolation were applied, ensuring continuity and reliability of the sensor signals. The numeric features underwent normalization to scale the values between 0 and 1, improving the model's convergence and stability during training.

Label encoding transformed the categorical activity labels into numerical classes compatible with machine learning algorithms. Furthermore, the preprocessed dataset retained subject IDs to facilitate a subject-wise train-test split, reflecting practical scenarios where models predict activities of unseen subjects.

Overall, this preprocessing framework ensures that clean, standardized, and properly labeled data is available for building accurate and generalizable activity recognition models.

C. Feature Selection

In this study, feature selection was carried out with a focus on maintaining a balance between model accuracy and energy efficiency, particularly important for deployment on mobile and wearable devices with limited battery life. Unlike conventional HAR approaches that use complex features derived from multiple sensors and various transformations,

our method specifically targets simple yet effective features that demand minimal computational power.

We limited our feature set to the statistical measures of mean and standard deviation calculated from the body acceleration signals. These features are known to capture essential movement patterns while avoiding the computational expense associated with other features such as jerk, frequency-domain components, or sensor magnitudes. By focusing exclusively on these key features, the resulting model benefits from reduced dimensionality, faster processing, and lower energy usage, which are crucial for real-time and continuous human activity recognition on embedded devices.

This carefully curated feature subset ensures that the model remains lightweight without sacrificing the ability to distinguish between different human activities effectively. Additionally, it simplifies the overall data pipeline, making the model more interpretable and easier to maintain.

D. Data Splitting

To ensure a reliable and fair evaluation of our Human Activity Recognition models, we performed data splitting at the subject level. This approach prevents data leakage and simulates a realistic scenario where the trained model encounters data from entirely new users during testing.

Specifically, the combined dataset containing all subjects' data was partitioned using a group-aware methodology. We utilized subject identifiers to ensure that all sensor recordings from a single individual reside exclusively within either the training or the testing subset. This is crucial in avoiding overly optimistic evaluation results that can occur if the same subject appears in both sets.

For the general model training, the dataset was split into 80% training and 20% test segments based on this subject-wise grouping. In addition, for personalized modeling, data corresponding to each subject was independently divided into personalized training and testing data subsets, enabling models to learn user-specific movement patterns and to be evaluated on unseen portions of the same user's data.

This stratified, subject-wise data splitting strategy helps maintain the integrity of the evaluation process while accurately reflecting the model's ability to generalize across different individuals.

E. Model Architecture

For the task of Human Activity Recognition (HAR), we selected the Light Gradient Boosting Machine (LightGBM) as our core machine learning algorithm. LightGBM is a gradient boosting framework that uses tree-based learning algorithms, known for its efficiency, fast training speed, and high accuracy. These characteristics make it suitable for real-time HAR deployment on resource-constrained mobile and wearable devices.

Algorithm Justification

LightGBM offers several advantages over traditional classifiers and deep learning models for this application:

- i. It requires less computational resources and memory compared to deep neural networks.
- ii. It naturally handles categorical and numerical data.
- iii. It provides fast inference, which is vital for battery-operated devices.

- iv. It supports parallel and GPU learning, expediting training and testing cycles.

Hyperparameters

The model was configured with the following main hyperparameters to balance between accuracy and computational efficiency:

- i. Number of estimators (trees): 100
- ii. Max depth of each tree: 6
- iii. Learning rate: 0.1
- iv. Subsampling rate: 0.8 (percentage of samples used for training each tree)
- v. Column sampling by tree: 0.8 (percentage of features selected per tree)
- vi. Random state: 42 (for reproducibility)

These hyperparameters were selected based on a combination of literature review and empirical tuning to avoid overfitting while maintaining good predictive performance.

General vs Personalized Pipeline Differences

We implemented two parallel modeling approaches:

- i. **General Model:** A single LightGBM classifier was trained on the aggregated data from all subjects combined. This model aims to learn generalized patterns of human activities across various individuals.
- ii. **Personalized Models:** Separate LightGBM models were trained for each subject using their own subset of data. Each subject's data was split into a personal train-test partition to capture their unique movement characteristics.

The personalized pipeline allows the model to adapt well to subject-specific variations, resulting in improved accuracy and responsiveness compared to the general model. However, it requires training and maintaining multiple models proportional to the number of users.

F. Training Procedure.

Model training was executed under two distinct scenarios: a pooled general model and multiple personalized models. For the general model, data from all subjects were combined to create a single training set. This unified dataset was used to fit a LightGBM classifier, enabling the model to learn broadly applicable patterns of human activity across multiple individuals.

In contrast, the personalized modeling approach involved training separate LightGBM classifiers for each subject. Here, the data belonging to each individual was split into personalized training and testing subsets. This dedicated training allows the model to capture individual-specific variations in movement and behavior, improving classification performance.

To ensure reliable model training, a minimum sample size threshold was enforced. Specifically, only subjects with at least 10 samples for training and 5 samples for testing were included in the personalized model training process. This criterion ensured that the models had sufficient data to learn

meaningful patterns while avoiding overfitting or unreliable predictions due to data scarcity.

Models were trained until performance convergence on the training data, with hyperparameters fixed as described previously. Regular evaluation on testing subsets verified the models' ability to generalize and informed any necessary refinements.

G. Evaluation Metrics .

In this study, preprocessing plays a critical role in preparing the raw sensor data for effective model training and evaluation. Initially, the separate training and testing datasets from the UCI HAR dataset were combined into a single dataset for unified processing. Each record in this dataset includes tri-axial accelerometer and gyroscope readings, subject identifiers, and activity labels corresponding to six common human activities.

To assess the performance of our Human Activity Recognition models, we utilized several widely accepted evaluation metrics that collectively provide a comprehensive view of classification quality.

Accuracy measures the overall proportion of correctly predicted activity instances among all tested samples. It gives a general sense of model correctness but may not fully reflect performance on imbalanced classes.

Precision quantifies the ratio of true positive predictions to the sum of true positives and false positives for each activity. It indicates how accurate the model's positive predictions are.

Recall (or sensitivity) expresses the ratio of true positive predictions to the total actual positives. This metric emphasizes the model's ability to capture all instances of a given activity.

F1-score is the harmonic mean of precision and recall, providing a balanced measure that accounts for both false positives and false negatives, particularly useful for class-imbalanced datasets.

These metrics were computed both overall—aggregating all activities and subjects—and individually per subject to analyze how well the models generalize across different individuals. Additionally, per-activity metrics were calculated to identify strengths and weaknesses in recognizing specific movements.

The combination of these evaluation strategies ensures a thorough and nuanced understanding of model behavior, guiding improvements and validating the effectiveness of our personalized and battery-efficient approach.

H. Figures and Tables

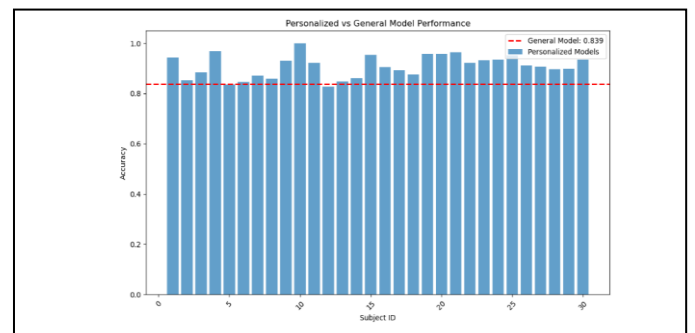
TABLE I.

ACTIVITY	PRECISION (MACRO AVG)	RECALL	F1- SCORE
LAYING	0.84	0.83	0.84
SITTING	0.90	0.81	0.85
STANDING	0.79	0.87	0.83
WALKING	0.85	0.77	0.81
WALKING_DOWNSTAIRS	0.90	0.90	0.90
WALKING_UPSTAIRS	0.78	0.85	0.81

Fig. 1. Classification Report (General model)

✔ LightGBM Accuracy: 0.8386				
📊 Classification Report:				
	precision	recall	f1-score	support
LAYING	0.84	0.83	0.84	395
SITTING	0.90	0.81	0.85	370
STANDING	0.79	0.87	0.83	388
WALKING	0.85	0.77	0.81	324
WALKING_DOWNSTAIRS	0.90	0.90	0.90	283
WALKING_UPSTAIRS	0.78	0.85	0.81	315
accuracy			0.84	2075
macro avg	0.84	0.84	0.84	2075
weighted avg	0.84	0.84	0.84	2075

Fig. 2. Personalized vs Model Performance



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